Cognitive Robotics Lab – 3

23BRS1177

Videos: https://github.com/Prayagraj27/Cognitive-Robotics-Lab

Traffic Lights (Stop at Red, Go at Green):

Code:

```
#!/usr/bin/env python3


import rclpy

from rclpy.node import Node


from sensor_msgs.msg import Image

from geometry_msgs.msg import Twist


from cv_bridge import CvBridge


import cv2

import numpy as np


class ColorDetector(Node):


    def __init__(self):

        super().__init__('color_detector')


        # Change this if your camera topic is different

        self.camera_topic = '/pi_camera/image_raw'


        # Robot movement topic

        self.cmd_vel_topic = '/cmd_vel_nav'
```

```
    self.bridge = CvBridge()

    self.image_subscriber = self.create_subscription(
        Image,
        self.camera_topic,
        self.image_callback,
        10
    )

    self.cmd_vel_publisher = self.create_publisher(
        Twist,
        self.cmd_vel_topic,
        10
    )

    self.get_logger().info('Color detector started')
    self.get_logger().info(
        f'Camera topic: {self.camera_topic}'
    )
    self.get_logger().info(
        f'Command topic: {self.cmd_vel_topic}'
    )

def image_callback(self, msg):

    try:
        frame = self.bridge.imgmsg_to_cv2(
            msg,
            desired_encoding='bgr8'
        )
    except Exception as e:
        self.get_logger().error(
            f'Could not convert image: {e}'
        )
```

```
    return

# Convert BGR image to HSV
hsv = cv2.cvtColor(frame, cv2.COLOR_BGR2HSV)

# ------------------------------
# GREEN
# ------------------------------
lower_green = np.array([35, 80, 50])
upper_green = np.array([90, 255, 255])

green_mask = cv2.inRange(
    hsv,
    lower_green,
    upper_green
)

# ------------------------------
# RED
# Red wraps around the HSV range,
# so we use two ranges.
# ------------------------------
lower_red1 = np.array([0, 100, 50])
upper_red1 = np.array([10, 255, 255])

lower_red2 = np.array([170, 100, 50])
upper_red2 = np.array([180, 255, 255])

red_mask1 = cv2.inRange(
    hsv,
    lower_red1,
    upper_red1
)

red_mask2 = cv2.inRange(
```

```
    hsv,
    lower_red2,
    upper_red2
)

red_mask = cv2.bitwise_or(
    red_mask1,
    red_mask2
)

# Count colored pixels
green_pixels = cv2.countNonZero(green_mask)
red_pixels = cv2.countNonZero(red_mask)

# Robot command
cmd = Twist()

# RED HAS PRIORITY
if red_pixels > 5000:

    cmd.linear.x = 0.0
    cmd.angular.z = 0.0

    self.get_logger().info(
        'RED detected -> STOP'
    )

elif green_pixels > 5000:

    cmd.linear.x = 0.15
    cmd.angular.z = 0.0

    self.get_logger().info(
        'GREEN detected -> MOVE'
    )
```

```
        else:

            cmd.linear.x = 0.0
            cmd.angular.z = 0.0

        self.cmd_vel_publisher.publish(cmd)

def main(args=None):

    rclpy.init(args=args)

    node = ColorDetector()

    try:
        rclpy.spin(node)

    except KeyboardInterrupt:
        pass

    # Make absolutely sure the robot stops
    stop_cmd = Twist()
    node.cmd_vel_publisher.publish(stop_cmd)

    node.destroy_node()

    rclpy.shutdown()

if __name__ == '__main__':
    main()
```

Outputs:

```
ssh: Could not resolve hostname veerobot: Temporary failure in name resolution
student@jazzy-jalisco:~$ ssh veerobot@192.168.0.127
The authenticity of host '192.168.0.127 (192.168.0.127)' can't be established.
ED25519 key fingerprint is SHA256:POF/SQdHVT0+O9qR1MgsZgDiATDCZH7ucZgeYLpQva8.
This host key is known by the following other names/addresses:
    ~/.ssh/known_hosts:1: [hashed name]
    ~/.ssh/known_hosts:4: [hashed name]
    ~/.ssh/known_hosts:5: [hashed name]
    ~/.ssh/known_hosts:6: [hashed name]
    ~/.ssh/known_hosts:7: [hashed name]
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.0.127' (ED25519) to the list of known hosts.
veerobot@192.168.0.127's password:
Welcome to Ubuntu 24.04.3 LTS (GNU/Linux 6.8.0-1044-raspi aarch64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

 System information as of Thu Aug 20 06:25:25 AM UTC 2026

  System load:  2.66               Temperature:            46.8 C
  Usage of /:   26.1% of 28.28GB   Processes:              164
  Memory usage: 6%                 Users logged in:        0
  Swap usage:   0%                 IPv4 address for wlan0: 192.168.0.127

 * Strictly confined Kubernetes makes edge and IoT secure. Learn how MicroK8s
   just raised the bar for easy, resilient and secure K8s cluster deployment.

   https://ubuntu.com/engage/secure-kubernetes-at-the-edge

This power supply is not capable of supplying 5A; power to peripherals
will be restricted

See man:pemmican-cli(1) for information on suppressing this warning,
or https://rptl.io/rpi5-power-supply-info for more information on the
Raspberry Pi 5 power supply

Expanded Security Maintenance for Applications is not enabled.

348 updates can be applied immediately.
To see these additional updates run: apt list --upgradable
```

```
veerobot@beetlebot-127:~$ echo $ROS_DISTRO
jazzy
veerobot@beetlebot-127:~$ source ~/beetlebot_ws/install/setup.bash
-bash: /home/veerobot/beetlebot_ws/install/setup.bash: No such file or directory
veerobot@beetlebot-127:~$ ros2 topic list
/battery_voltage
/cmd_vel
/cmd_vel_joy
/diagnostics
/imu/data
/imu/data_raw
/joint_states
/joy
/joy/set_feedback
/lyra/armed
/odometry/filtered
/parameter_events
/robot_description
/rosout
/set_pose
/tf
/tf_static
/wheel/odom
/wheel_rpm
/wheel_ticks
veerobot@beetlebot-127:~$ source ~/beetlebot_ws/install/setup.bash
-bash: /home/veerobot/beetlebot_ws/install/setup.bash: No such file or directory
veerobot@beetlebot-127:~$ export ROS_DOMAIN_ID = 127
-bash: export: `=': not a valid identifier
-bash: export: `127': not a valid identifier
veerobot@beetlebot-127:~$  lyra-help
╔═══════════════════════════════════════════════════════╗
                    LYRA ROBOT COMMAND UTILITY
╚═══════════════════════════════════════════════════════╝

LAUNCH COMMANDS:
  lyra-launch-base              Launch base robot only
```

```
/wheel_ticks
student@jazzy-jalisco:~/beetlebot_ws$ ros2 run camera_ros camera_node
Package 'camera_ros' not found
student@jazzy-jalisco:~/beetlebot_ws$ ros2 service call /lyra/arm std_srvs/srv/Trigger
waiting for service to become available...
requester: making request: std_srvs.srv.Trigger_Request()

response:
std_srvs.srv.Trigger_Response(success=True, message='ARM command sent')

student@jazzy-jalisco:~/beetlebot_ws$ ros2 service call /lyra/arm std_srvs/srv/Trigger
waiting for service to become available...
requester: making request: std_srvs.srv.Trigger_Request()

response:
std_srvs.srv.Trigger_Response(success=True, message='ARM command sent')

student@jazzy-jalisco:~/beetlebot_ws$ ros2 topic echo /cmd_vel_nav
^Cstudent@jazzy-jalisco:~/beetlebot_ws$ ros2 topic echo /cmd_vel_nav
^Cstudent@jazzy-jalisco:~/beetlebot_ws$
student@jazzy-jalisco:~/beetlebot_ws$ ros2 service call /lyra/arm std_srvs/srv/Trigger
waiting for service to become available...
requester: making request: std_srvs.srv.Trigger_Request()

response:
std_srvs.srv.Trigger_Response(success=True, message='ARM command sent')

student@jazzy-jalisco:~/beetlebot_ws$ ros2 topic echo /cmd_vel_nav
^Cstudent@jazzy-jalisco:~/beetlebot_ws$ ros2 topic info /cmd_vel_nav
Type: geometry_msgs/msg/Twist
Publisher count: 0
Subscription count: 2
student@jazzy-jalisco:~/beetlebot_ws$ cd ~/beetlebot_ws
colcon build --symlink-install --packages-select color_detector
Starting >>> color_detector
Finished <<< color_detector [0.86s]

Summary: 1 package finished [0.95s]
student@jazzy-jalisco:~/beetlebot_ws$ source ~/beetlebot_ws/install/setup.bash
student@jazzy-jalisco:~/beetlebot_ws$ ros2 pkg executables color_detector
color_detector color_detector
```

Sign Detection Outputs:

```
student@jazzy-jalisco:~/beetlebot_ws$ source ~/beetlebot_ws/install/setup.bash
student@jazzy-jalisco:~/beetlebot_ws$ export ROS_DOMAIN_ID=127
student@jazzy-jalisco:~/beetlebot_ws$ ros2 pkg create --build-type ament_python color_detector --dependencies rclpy sensor_msgs geometry_ms
going to create a new package
package name: color_detector
destination directory: /home/student/beetlebot_ws
package format: 3
version: 0.0.0
description: TODO: Package description
maintainer: ['student <student@todo.todo>']
licenses: ['TODO: License declaration']
build type: ament_python
dependencies: ['rclpy', 'sensor_msgs', 'geometry_msgs', 'cv_bridge']
creating folder ./color_detector
creating ./color_detector/package.xml
creating source folder
creating folder ./color_detector/color_detector
creating ./color_detector/setup.py
creating ./color_detector/setup.cfg
creating folder ./color_detector/resource
creating ./color_detector/resource/color_detector
creating ./color_detector/color_detector/__init__.py
creating folder ./color_detector/test
creating ./color_detector/test/test_copyright.py
creating ./color_detector/test/test_flake8.py
creating ./color_detector/test/test_pep257.py

[WARNING]: Unknown license 'TODO: License declaration'.  This has been set in the package.xml, but no LICENSE file has been created.
It is recommended to use one of the ament license identifiers:
Apache-2.0
BSL-1.0
BSD-2.0
BSD-2-Clause
BSD-3-Clause
GPL-3.0-only
LGPL-3.0-only
MIT
MIT-0
```

```
student@jazzy-jalisco:~/beetlebot_ws/src$ dir
color_detector
student@jazzy-jalisco:~/beetlebot_ws/src$ cd color_detector/
student@jazzy-jalisco:~/beetlebot_ws/src/color_detector$ ls
color_detector  package.xml  resource  setup.cfg  setup.py  test
student@jazzy-jalisco:~/beetlebot_ws/src/color_detector$ cd color_detector/
student@jazzy-jalisco:~/beetlebot_ws/src/color_detector/color_detector$ nano color_detector.py
student@jazzy-jalisco:~/beetlebot_ws/src/color_detector/color_detector$ nano ~/beetlebot_ws/src/color_detector/color_detector/color_detector.py
student@jazzy-jalisco:~/beetlebot_ws/src/color_detector/color_detector$ chmod +x ~/beetlebot_ws/src/color_detector/color_detector/color_detector.py
student@jazzy-jalisco:~/beetlebot_ws/src/color_detector/color_detector$ cd ..
student@jazzy-jalisco:~/beetlebot_ws/src/color_detector$ nano setup.py
student@jazzy-jalisco:~/beetlebot_ws/src/color_detector$ colcon build --packages-select color_detector
Starting >>> color_detector
Finished <<< color_detector [0.94s]

Summary: 1 package finished [1.05s]
student@jazzy-jalisco:~/beetlebot_ws/src/color_detector$ source ~/beetlebot_ws/install/setup.bash
student@jazzy-jalisco:~/beetlebot_ws/src/color_detector$ ros2 pkg executables color_detector
Package 'color_detector' not found
student@jazzy-jalisco:~/beetlebot_ws/src/color_detector$ ls
build  color_detector  install  log  package.xml  resource  setup.cfg  setup.py  test
student@jazzy-jalisco:~/beetlebot_ws/src/color_detector$ cd color_detector/
student@jazzy-jalisco:~/beetlebot_ws/src/color_detector/color_detector$ ros2 pkg executables color_detector
Package 'color_detector' not found
student@jazzy-jalisco:~/beetlebot_ws/src/color_detector/color_detector$ cd ..
student@jazzy-jalisco:~/beetlebot_ws/src/color_detector$ ls ~/beetlebot_ws/install/color_detector
ls: cannot access '/home/student/beetlebot_ws/install/color_detector': No such file or directory
student@jazzy-jalisco:~/beetlebot_ws/src/color_detector$ cd ~/beetlebot_ws
rm -rf build install log
colcon build --symlink-install
Starting >>> color_detector
Finished <<< color_detector [0.88s]

Summary: 1 package finished [0.97s]
student@jazzy-jalisco:~/beetlebot_ws$ source ~/beetlebot_ws/install/setup.bash
student@jazzy-jalisco:~/beetlebot_ws$ ros2 pkg list | grep color
color_detector
student@jazzy-jalisco:~/beetlebot_ws$ ros2 pkg executables color_detector
color_detector color_detector
student@jazzy-jalisco:~/beetlebot_ws$ ros2 run color_detector color_detector
```